Do You Need the Big Model? Choosing Between Llama 3 8B and 70B

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1.Abstract

Developers often choose the largest AI model they can afford, even when a smaller one may be more convenient. This guide compares Llama 3 8B and Llama 3 70B, which share the same architecture and differ mainly in size, to show when the smaller model is good enough. With the help of published benchmark scores, model config files, and an energy study, I have compared the quality of replies of knowledge, code, and math and reasoning prompts, plus memory and energy use. 70B scores higher on every task, but 8B needs about 16 GB of memory against 140 GB, and in a particular study, 70B used about 5.8 times more energy. I have used the results to form a six-question decision flowchart. The main advice is to start with 8B and move to 70B only if the task needs it.

2. Method

I compared the Instruct versions of both models, because that is the version most people use. I compared them on three kinds of tasks: knowledge questions, code, and math and reasoning.
I did not run new experiments. I used published benchmark scores from Meta's model card. Every number is logged in my spreadsheet with its link and the date I accessed it.
Memory figures are calculated from each model's config files. Energy figures come from a published study, and I show Pakistan's grid carbon intensity as a range because sources disagree.

3. Results by task

These are the scores Meta reports for the Instruct models. The gap is the 70B scores minus the 8B scores.
Knowledge (MMLU, 5-shot): 8B scores 68.4 and 70B scores 82.0, a gap of 13.6 points. 8B gets about 32% of questions wrong, against 18% for 70B.
Code (HumanEval, 0-shot): 8B scores 62.2 and 70B scores 81.7, a gap of 19.5 points. 8B fails on about 38% of problems, against 18% for 70B.
Math, easier (GSM-8K, 8-shot): 8B scores 79.6 and 70B scores 93.0, a gap of 13.4 points. 8B fails about 2.9 times as often as 70B (20.4% against 7.0%).
Math, harder (MATH, 4-shot): 8B scores 30.0 and 70B scores 50.4, a gap of 20.4 points. Both models struggle.
Reasoning (GPQA, 0-shot): 8B scores 34.2 and 70B scores 39.5, a gap of 5.3 points. Both models score low.
A gap of 13 to 20 points can mean 8B fails about twice as often as 70B, which matters more for a real product than the raw score does.

3.1 Knowledge and Q&A

8B is a reasonable fit for general question answering where an occasional miss is acceptable or the answers can be checked. 70B is worth it when wrong answers are costly, such as in medical, legal, or financial settings.

3.2 Code

8B can handle short, well-specified functions and autocomplete-style help. 70B is the better choice when the code has to be right on the first attempt, or the task is longer or loosely specified. HumanEval only tests short standalone functions, so it says little about large codebases and thus result may not be accurate.

3.3 Math and reasoning

8B is usable on simple word problems where results can be verified, but it fails much more often than 70B. On harder math or graduate-level science questions, neither model is reliable without checking. 70B is the safer choice, though on GPQA the advantage is small.

4. Memory and hardware (technical)

Both models have 8 KV heads and an 8,192-token context. 8B has 32 layers and 70B has 80. At 16-bit precision, each parameter takes 2 bytes of memory. So 8B needs about 16 GB for its weights and 70B needs about 140 GB. 16 GB fits on one consumer GPU. 140 GB does not fit on a single 80 GB GPU, so you need at least two.
The model also keeps a memory of the conversation so far, called the KV cache, and it grows with every token. From the config files, I calculate 128 KB per token for 8B and 320 KB per token for 70B. At 8,192 tokens that is about 1 GB for 8B and 2.5 GB for 70B, for a single conversation. This adds up when many people use the model at the same time.
In short, 8B fits on one affordable GPU, and 70B needs data-center style hardware.

5. Energy and environmental impact:

Training first. Meta reports about 1.3 million GPU hours for Llama 3 8B and 6.4 million for 70B. A GPU is the powerful chip that runs AI models. One GPU running nonstop would need about 148 years to train the 8B model and about 730 years to train the 70B model, which is about 4.9 times more. Estimated training emissions are 390 tonnes of CO2 for 8B and 1,900 tonnes for 70B.

Running the models shows a similar gap. I found one study that ran both models on the same task, processing 175 clinical documents on A100 GPUs, which are data-center chips. 8B used 0.14 kWh and 70B used 0.81 kWh, and 70B needed two 80 GB GPUs. A kWh is the unit on an electricity bill. A 100-watt light bulb running for 10 hours uses 1 kWh.

Scaled to 1,000 documents, 8B uses about 0.80 kWh, like a light bulb left on for 8 hours. 70B uses about 4.63 kWh, like the same bulb left on for nearly two days. So 70B used about 5.8

times more energy.

Pakistan's grid produces between about 245 and 347 g of CO2 per kWh, depending on the source. Using that range, choosing 8B saves about 0.9 to 1.3 kg of CO2 per 1,000 documents. That is small for one user. At 1 million documents it adds up to about 3,800 kWh, enough to keep a light bulb on for more than four years, and about 0.9 to 1.3 tonnes of CO2.

These figures come from one study, one task, and one type of GPU. Other work shows that longer prompts raise energy use per token, so a different workload could give different numbers. The savings only count if 8B is good enough for the task, which is why the quality results come first.

6. Decision flowchart

Go through these questions in order and stop at the first result you reach.
1. Would a wrong answer be costly, like in medical, legal, or money decisions? If yes, use 70B and check the answers personally. MMLU shows that 8B gets 31.6% of questions wrong and 70B gets 18.0% wrong.
2. Is the task hard multi-step reasoning or advanced math? If yes, use 70B and still check the outputs. On MATH, 8B scores 30.0 and 70B scores 50.4.
3. Does the code have to be right on the first try? If yes, use 70B. On HumanEval, 8B scores 62.2 and 70B scores 81.7.
4. Can you run 70B hardware? If no, use 8B. 70B needs about 140 GB of memory at 16-bit, which is at least two 80 GB GPUs. 8B needs about 16 GB and fits on one GPU.
5. Will you use it a lot, or is saving energy a priority? If yes, make 8B your starting point and consider using 70B only for the hard requests. In the study I found, 70B used about 5.8 times more energy than 8B, which is about 3.8 kWh extra per 1,000 documents. Using 70B only for hard requests is an estimate because I have not tested this.
6. Test both models on 20 to 50 examples from your own task. If 8B does well enough, use 8B. If it doesn't, use 70B.
Summary of the steps (teal means use 8B, coral means use 70B):

Step	Question	If yes	If no
1	Would a wrong answer be costly, like in medical, legal, or money decisions?	Use 70B and check the answers personally	Go to step 2
2	Is the task hard multi-step reasoning or advanced math?	Use 70B and still check the outputs	Go to step 3
3	Does the code have to be right on the first try?	Use 70B	Go to step 4
4	Can you run 70B	Go to step 5	Use 8B

	hardware?		
5	Will you use it a lot, or is saving energy a priority?	Make 8B your starting point and consider 70B only for hard requests (untested). Go to step 6	Go to step 6
6	Test both models on 20 to 50 examples from your own task. Does 8B do well enough?	Use 8B	Use 70B

7. Limitations

This guide uses third party published numbers as I did not run my own tests, so results may vary.
The quality scores are Meta's own results on specific prompts and information. these Benchmarks are variable, so your task may behave differently.
The energy figures come from one study on one task and one type of GPU. Longer prompts, different hardware, or a different batch size could change the results.
Pakistan's grid carbon intensity differs between sources and between seasons, so the CO2 savings are a range and not an exact figure.
I only compared Llama 3 8B and 70B. Newer or different models may not follow the same pattern.
The decision questions are my own observation or logical thinking, not tested cutoffs.
The guide deals with only a few types of prompts so your prompt type may not be mentioned here

8. Conclusion

Bigger is not always worth it. With the benchmarks I used, 70B beat 8B on every task, but the gap changes a lot from task to task. 8B is much cheaper to run, about 16 GB of memory against 140 GB, and in the one study I found, 70B used about 5.8 times more energy.
So the default should be to start with 8B and move to 70B only if your task needs it, such as when mistakes are costly, the reasoning is hard, or the code has to be right the first time. Test both models on 20 to 50 of your own examples before you decide.
When 8B is good enough, choosing it saves about 0.9 to 1.3 kg of CO2 per 1,000 documents on Pakistan's grid. If the other factors are negligible 8B should always be used to reduce environmental effect.

References

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